

Joongmin Shin

Senior Researcher, Human-Inspired AI Research | Founder & Lead at KUDoc

Seoul, Republic of Korea

✉ tlswndals13@korea.ac.kr | 🏠 [shinjm-maker.github.io](https://github.com/shinjm-maker) | [in](#) [shinjm-maker](#) | [🎓 Google Scholar](#)

Research Interests

I am a Senior Researcher at the Human-Inspired AI Research Institute, Korea University, where I work with **Prof. Heuseok Lim** and **Prof. Jaehyung Seo** and lead **KUDoc**. I received my B.S. in Information and Computer Engineering and my M.S. in Artificial Intelligence (Advisor: Prof. Hyuk-Chul Kwon) from Pusan National University.

My research uses **structure-based reasoning** to address **real-world and industrial problems**, with a focus on making AI genuinely useful to people. My master's research in parsing gave me this structural perspective, while my later work in human-inspired AI clarified how that perspective could serve practical human needs. I have since developed a coherent research program spanning multimodal and agentic AI.

Looking ahead, I aim to study how AI systems should represent, select, update, and verify information so that they remain reliable as they reason over complex multimodal inputs and engage in extended interactions. Over the longer term, I am interested in extending these principles to world models and embodied agents in robotics.

My research has yielded **14 publications**, including **6 first-author papers** at **NeurIPS**, **ACL**, **CVPR**, and **EMNLP**; these include main-track papers at all four venues, with my ACL 2026 paper selected for **oral presentation**. I also hold **5 patents** and have received **7 awards**, including **3 paper awards**. KUDoc-affiliated projects have produced **15 publications**, **3 technology transfers**, and multiple industry collaborations.

Structure-Based Reasoning	Parsing · Hierarchical & Knowledge Graphs
Multimodality	Documents · Video · World Models
Agentic AI	Memory · Verification

Selected Publications

ADMIT: Support-Gated Memory-Write Admission for Document QA Agents

* First author or co-first author

Joongmin Shin*, Gyuho Shim, Hyeonseok Moon, Jaehyung Seo
NeurIPS 2026 – Conference on Neural Information Processing Systems (Main)

PILAR: A Page-Grounded Unified Evidence Representation via an Entity-Linked Assertion Graph for Open-Domain QA Agents over Multimodal Document Corpora

Joongmin Shin*, Gyuho Shim, Jung-Hun Lee, Jaehyung Seo
EMNLP 2026 – Conference on Empirical Methods in Natural Language Processing (Findings)

HiKEY: Hierarchical Multimodal Retrieval for Open-Domain Document Question Answering

Joongmin Shin*, Gyuho Shim, Jeongbae Park, Jaehyung Seo, Heuseok Lim
ACL 2026 – Annual Meeting of the Association for Computational Linguistics (Main, Oral)

M3DocDep: Multi-modal, Multi-page, Multi-document Dependency Chunking with Large Vision-Language Models

Joongmin Shin*, Jeongbae Park, Jaehyung Seo, Heuseok Lim
CVPR 2026 – IEEE/CVF Conference on Computer Vision and Pattern Recognition (Main)

MultiDocFusion: Hierarchical and Multimodal Chunking Pipeline for Enhanced RAG on Long Industrial Documents

Joongmin Shin*, Chanjun Park, Jeongbae Park, Jaehyung Seo, Heuseok Lim
EMNLP 2025 – Conference on Empirical Methods in Natural Language Processing (Main)

Intelligent Predictive Maintenance RAG Framework for Power Plants: Enhancing QA with StyleDFS and Domain Specific Instruction Tuning

Seongtae Hong*, Joongmin Shin*, Jaehyung Seo, Taemin Lee, Jeongbae Park, Cho Man Young, Byeongho Choi, Heuseok Lim
EMNLP 2024 – Conference on Empirical Methods in Natural Language Processing (Industry Track)

Search-Based Generation Techniques for Improving LLM Responses: A Comparative Study of Zero-shot and RAG on GPT-3.5 and GPT-4

Joongmin Shin*, SeongReol Park, Jung-Hun Lee
KIICE 2023 – Korea Institute of Information and Communication Engineering – Outstanding Paper Award

EDT5: Proposed Embedding Model of T5 Encoder-Decoder Structure

Joongmin Shin*, Jugyung Jung, Jung-Hun Lee, Sanghyun Cho, Minh Kim, Miyeon Kim, Hyuk-Chul Kwon
KSC 2022 – Korea Software Congress – Outstanding Paper Award

A Dependency Parsing Model with Reinforced Head-Dependent Constraint Rules: Combining Deep Learning and Linguistic Knowledge

Joongmin Shin*, Hyuk-Chul Kwon
HCLT 2022 – Conference on Human and Cognitive Language Technology – Student Research Award

Research Experience

KUDoc: Korea University Document AI Research Group

Seoul, Republic of Korea

FOUNDER & LEAD

2024 – Present

- **Founding & research agenda:** Built the group from the ground up as an independent line of document AI work at Korea University.
- **Project coordination:** Oversaw KUDoc-affiliated projects that produced 15 publications, 3 technology transfers, and multiple industry collaborations.
- **Technical direction:** Steered research across multimodal document parsing, structure-aware retrieval, grounded document QA, and agentic systems.

Human-Inspired AI Research Institute, Korea University

Seoul, Republic of Korea

SENIOR RESEARCHER

Jun 2023 – Present

- **Structure-based multimodal reasoning:** Developed retrieval and reasoning methods that preserve document structure – layout, tables, figures, and cross-page dependencies – turning noisy multi-page documents and video into grounded evidence for QA.
- **First-author work:** Led first-author papers at NeurIPS, ACL, CVPR, and EMNLP on agent memory-write admission, page-grounded evidence graphs, hierarchical retrieval, multimodal document dependency parsing, and structure-preserving chunking.
- **Hybrid reasoning systems:** Built QA pipelines combining LLMs/LVLMs, knowledge graphs, symbolic rules, and evidence filtering for accurate, interpretable answers.
- **Applied validation:** Validated multimodal RAG in insurance, manufacturing, and healthcare, including a Samsung Fire & Marine insurance QA MVP that reduced manual document-search time by approximately 40%.

AI Lab, Pusan National University

Busan, Republic of Korea

GRADUATE RESEARCHER

Sep 2020 – Feb 2023

- **Korean dependency parsing:** Developed neural-symbolic dependency parsing methods that integrate morphological and syntactic information for robust Korean language understanding.
- **Table and document QA:** Designed pipelines that convert semi-structured tables and documents into structured representations for question answering and interpretable reasoning.
- **Applied collaborations:** Contributed to Korean TTS prosody prediction with KT AI2XL and additional projects in finance QA, sentiment-based marketing, and recommendation systems.

Education

Pusan National University

Busan, Republic of Korea

M.S. IN ARTIFICIAL INTELLIGENCE

Mar 2021 – Feb 2023

- **Advisor:** Prof. Hyuk-Chul Kwon | AI Research Lab
- **Thesis:** *A Neural-Symbolic Model for Overcoming Deep Learning Limitations in Korean Dependency Parsing*
- **Selected outcomes:** Developed neural-symbolic Korean dependency parsing and table/document QA methods; received Outstanding Paper and Student-led Research Project awards.

Pusan National University

Busan, Republic of Korea

B.S. IN INFORMATION AND COMPUTER ENGINEERING

Mar 2016 – Feb 2021

- **Undergraduate Research:** Prof. Jeongu Kim’s group and Prof. Hyuk-Chul Kwon’s AI Research Lab
- **Selected outcomes:** Worked on knowledge-graph-based Korean NLP, Korean grammar correction, Korean TTS, and transformer-based Korean language models; received a merit-based Milestone Scholarship.

Selected Honors & Awards

2024	Deputy Prime Minister and Minister of Education Award , Competition for Best Practices in Academia–Industry Cooperation	NRF
2023	Outstanding Paper Award , Korea Institute of Information and Communication Engineering	KIICE
2023	Grand Prize , Samsung Fire & Marine Insurance LLM-based MVP Model Idea Competition	Samsung
2023	Excellence Award (Technology Track) , KU-Grant Program	Korea University
2022	Outstanding Paper Award , Korea Software Congress (KSC)	KSC
2022	Outstanding Research Award , Student-led Research Project, Graduate School of AI	Pusan National University
2020	Milestone Scholarship (Grade A) , SW-centered University Milestone Scholarship (merit-based)	Pusan National University

Patents

2026	Hierarchical Multimodal Retrieval for Open-Domain Document QA	Korea, Pending
2026	Multimodal Multi-Document Dependency Chunking	Korea, Pending
2025	Document Chunking and QA System	Korea · DP-2025-0093, Pending
Feb 2023	Rule Filtering for Neural Knowledge Inference	Korea · P2022-0340-KR00, Registered
Nov 2022	Adaptive Problem Recommendation	Korea · 10-2022-0068075, Registered

Technical Skills

Research	Multimodal LLMs, RAG, hierarchical retrieval, knowledge graphs, neural-symbolic reasoning
Programming & ML	Python, C/C++, PyTorch, Hugging Face, PEFT/LoRA
Retrieval & Evaluation	Elasticsearch, FAISS, lm-eval-harness
Languages	Korean (native), English (professional working proficiency)

Teaching & Mentoring

KUDoc, Korea University

Seoul, Republic of Korea

RESEARCH MENTOR

2024 – Present

- Mentor 3 M.S. students and 2 Ph.D. students on document AI, multimodal RAG, and structured evidence reasoning.
- Mentored research that led to 15 publications and 4 manuscripts currently under review at top-tier venues.

Pusan National University

Busan, Republic of Korea

TEACHING ASSISTANT & RESEARCH MENTOR

2019 – 2022

- Mentored undergraduate research teams in dependency parsing and Table QA; one team achieved SOTA on KLUE-DP.
- TA for Software Design, Web Application Programming, and Programming Principles; also mentored 100+ students during the PNU Software Hackathon.

Selected Research Projects

MULTIMODAL REASONING & DOCUMENT AI

Video Curation System – Multimodal Video QA

LEAD RESEARCHER · KOREA UNIVERSITY

Jul 2025 – Present

Video QA and RAG system for structured curation and grounded retrieval over long multimodal content.

DocGraph Copilot – Structure-Aware Multimodal RAG

LEAD RESEARCHER · KOREA UNIVERSITY

Jan 2025 – Present

Structure-aware multimodal chunking pipeline for grounded retrieval and long-document QA; produced first-author papers at CVPR 2026 and EMNLP 2025.

Science-Domain Korean LLM & RAG Pipeline

LEAD RESEARCHER · KOREA UNIVERSITY

Feb 2024 – Dec 2024

Korean LLM and RAG pipeline specialized for the science domain; led data preprocessing and chunking methodology development; produced an EMNLP 2024 Industry Track publication.

Synerpeace Pilot – Insurance Document Assistant

LEAD RESEARCHER · KOREA UNIVERSITY · SAMSUNG FIRE & MARINE INSURANCE

Jul 2023 – Jan 2024

Insurance document QA pilot combining LLM-based retrieval with production-oriented application development.

FOUNDATION MODELS AND ADAPTATION

National Representative K-AI Research Team

RESEARCHER · KOREA UNIVERSITY · NC AI CONSORTIUM

Aug 2025 – Jan 2026

Contributed to multilingual foundation-model development across data pipelines, post-training, and evaluation.

Foundation-Model Adaptation

RESEARCHER · KOREA UNIVERSITY · PUSAN NATIONAL UNIVERSITY · KT

Dec 2021 – Mar 2024

Contributed to Mi:deum K, KULLM, and Ko-Gemma across Korean adaptation, instruction tuning, and evaluation.

Robot Process Automation – PLC Control-Code Generation

RESEARCHER · KOREA UNIVERSITY · HYUNDAI MOBIS

Sep 2023 – Feb 2024

GPT-based assistant that generates PLC control code for smart-factory robot process automation, in collaboration with Hyundai Mobis.

Exobrain WiseQA – National QA

GRADUATE RESEARCHER · PUSAN NATIONAL UNIVERSITY · ETRI

Sep 2020 – Nov 2022

Table QA and neural-symbolic Korean sentence analysis for the national Exobrain project; introduced an Adapter-based architecture unifying separate text/table embedding models, reducing QA model parameters by approximately 50%.

All Publications

* First author or co-first author

INTERNATIONAL CONFERENCE PUBLICATIONS

ADMIT: Support-Gated Memory-Write Admission for Document QA Agents

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JOURNAL PUBLICATIONS

Multi-Paragraph Machine Reading Comprehension with Hybrid Reader over Tables and Text

Sanghyun Cho*, SeongReol Park, Hye-Lynn Kim, Jung-Hun Lee, Joongmin Shin, Hyuk-Chul Kwon

Applied Artificial Intelligence 38.1 (2024), p. 2367820. Taylor & Francis

Distance Based Korean WordNet (alias. KorLex) Embedding Model

SeongReol Park*, Joongmin Shin, Sanghyun Cho, Hyuk-Chul Kwon, Jung-Hun Lee

Applied Artificial Intelligence 38.1 (2024), p. 2398920. Taylor & Francis

DOMESTIC CONFERENCE PUBLICATIONS

Search-Based Generation Techniques for Improving LLM Responses: A Comparative Study of Zero-shot and RAG on GPT-3.5 and GPT-4

Joongmin Shin*, SeongReol Park, Jung-Hun Lee

KIICE 2023 – Korea Institute of Information and Communication Engineering – **Outstanding Paper Award**

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KSC 2022 – Korea Software Congress – **Outstanding Paper Award**

Evaluation of Generalization Performance in Korean Table Machine Reading Comprehension across Domain-Specific Datasets

HyeLin Kim*, Sanghyun Cho, Joongmin Shin, Hyuk-Chul Kwon

KSC 2022 – Korea Software Congress

A Dependency Parsing Model with Reinforced Head-Dependent Constraint Rules: Combining Deep Learning and Linguistic Knowledge

Joongmin Shin*, Hyuk-Chul Kwon

HCLT 2022 – Conference on Human and Cognitive Language Technology – **Student Research Award**

Rule-Augmented Neural Network-Based Dependency Parsing

Joongmin Shin*, Sanghyun Cho, Bongwoo Nam, Hyuk-Chul Kwon

KCC 2022 – Korea Computer Congress

Korean Machine Reading Comprehension Using Continual Learning

Joongmin Shin*, Sanghyun Cho, Hyuk-Chul Kwon

HCLT 2021 – Conference on Human and Cognitive Language Technology

MASTER’S THESIS

A Neural-Symbolic Model for Overcoming Deep Learning Limitations in Korean Dependency Parsing

Joongmin Shin*, Hyuk-Chul Kwon

Master’s Thesis, Pusan National University, 2023

First-Author Manuscripts Under Review

- Evaluating the practical utility of agentic memory — under review at **ICLR 2027**.
- Citation grounding for video question answering — under review at **ICLR 2027**.
- Evidence auditing and support-sensitive evaluation for multimodal QA — under review at **TACL**.
- Error-propagation diagnostics for PDF-to-RAG pipelines — under review at **IEEE TPAMI**.

First-Author Manuscripts in Preparation

- Neuro-symbolic world modeling — in preparation for **CVPR 2027**.